

SALEM-LECKRONE, IL, USA

WMO: 724330

Lat: 38.650N

Lon: 88.967W

Elev: 174

StdP: 99.26

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 03879

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.3	-10.9	-18.9	0.7	-12.1	-16.3	0.9	-9.7	11.4	5.6	10.5	3.5	2.8	310	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	34.1	24.2	32.8	24.2	31.9	23.9	26.5	31.5	25.7	30.7	25.0	29.9	3.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.6	29.4	24.1	19.4	28.8	23.4	18.6	27.8	83.4	31.6	80.2	30.6	76.8	30.1	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.5	7.5	-17.5	35.7	3.1	1.5	-19.8	36.8	-21.6	37.6	-23.3	38.5	-25.6	39.6		
(5)			-17.9	27.7	2.9	0.9	-20.1	28.3	-21.8	28.8	-23.4	29.3	-25.5	29.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.5	0.0	1.7	7.8	13.8	19.2	24.0	25.4	24.3	20.6	14.3	7.8	2.7		
	DBStd	10.19	6.36	6.15	6.06	5.23	4.70	3.23	2.86	2.96	3.93	5.32	5.38	5.58		
	HDD10.0	1051	316	241	115	23	1	0	0	0	0	17	104	234		
	HDD18.3	2543	570	466	328	151	51	2	1	2	21	150	318	484		
	CDD10.0	2332	4	8	48	137	284	419	476	443	317	149	37	8		
	CDD18.3	783	0	0	3	16	76	171	218	187	88	24	1	0		
	CDH23.3	7387	0	0	16	120	596	1624	2236	1764	812	212	6	0		
	CDH26.7	2794	0	0	1	13	167	632	945	697	283	57	1	0		
(14) Wind	WSAvg	3.0	3.9	3.8	3.8	3.8	3.0	2.5	1.9	1.7	1.8	2.7	3.4	3.6		
(15) Precipitation	PrecAvg	1088	83	61	86	102	137	108	95	81	78	84	100	72		
	PrecMax	1453	198	115	196	243	347	231	220	182	199	176	198	210		
	PrecMin	882	13	15	23	36	24	19	42	25	5	30	14	31		
	PrecStd	159	49	30	40	55	84	51	46	39	47	45	49	41		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.9	20.6	25.5	28.0	32.1	34.9	36.2	35.5	34.1	31.3	23.8	19.0		
		MCWB	15.2	15.8	16.8	18.7	22.5	23.7	24.1	24.3	23.0	21.8	16.3	15.4		
	2%	DB	15.2	17.4	22.5	26.4	30.2	33.2	34.1	33.5	32.2	27.9	21.2	16.2		
		MCWB	12.7	13.9	15.8	17.8	21.4	23.7	24.8	24.2	23.1	19.7	15.3	13.2		
	5%	DB	12.6	14.1	19.9	24.7	28.7	32.2	32.7	32.2	30.1	25.8	18.7	13.5		
		MCWB	10.1	9.9	14.5	17.0	20.9	23.4	25.1	24.0	22.4	18.1	14.1	11.0		
	10%	DB	9.5	11.3	17.7	22.7	27.1	30.5	31.4	30.8	27.8	23.0	16.7	11.0		
		MCWB	7.1	8.0	13.4	16.3	20.2	22.6	24.3	23.4	20.7	17.5	12.3	8.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.8	17.1	18.5	20.8	24.1	26.1	27.5	27.1	25.7	23.2	18.9	16.3		
		MCDB	17.4	19.5	23.1	25.8	29.0	31.5	31.8	31.4	30.8	28.7	22.1	18.0		
	2%	WB	13.5	14.3	17.0	19.5	23.1	25.2	26.8	26.1	24.4	21.8	16.9	13.9		
		MCDB	15.1	17.1	20.9	24.7	27.9	31.2	31.6	31.2	29.3	26.2	19.4	15.8		
	5%	WB	10.5	11.4	15.4	18.3	22.1	24.5	26.1	25.3	23.4	20.4	15.1	11.5		
		MCDB	12.2	13.5	19.1	22.8	26.7	30.0	31.0	30.6	28.1	23.9	18.5	13.2		
	10%	WB	6.6	7.9	13.9	17.1	21.1	23.7	25.4	24.4	22.4	18.4	12.7	8.8		
		MCDB	9.4	10.9	17.4	21.3	25.3	28.5	29.9	28.9	26.6	22.1	16.3	10.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	8.5	9.6	10.7	9.9	10.3	10.0	10.8	11.9	11.4	9.5	7.4		
		MCDBR	11.4	12.5	12.1	12.6	11.4	11.6	11.1	12.2	13.3	13.5	12.6	10.3		
	5% WB	MCWBR	9.4	9.6	7.6	6.8	5.1	4.4	4.3	5.0	5.5	6.2	8.1	8.4		
		MCWBR	10.6	11.6	10.9	10.6	9.9	10.1	9.8	10.4	11.1	11.1	10.8	9.3		
(40) Clear-Sky Solar Irradiance	taub	taub	0.307	0.321	0.361	0.399	0.435	0.436	0.463	0.461	0.409	0.357	0.318	0.305		
		taud	2.457	2.404	2.359	2.254	2.210	2.234	2.160	2.153	2.288	2.433	2.488	2.479		
	Ebn at Noon	Ebn at Noon	871	904	897	878	849	847	819	810	834	851	851	847		
		Edh at Noon	81	97	113	134	142	139	149	146	120	94	78	74		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.97	2.75	3.72	4.92	5.52	6.39	6.22	5.62	4.70	3.39	2.28	1.67		
	RadStd	RadStd	0.20	0.29	0.35	0.28	0.48	0.42	0.41	0.32	0.32	0.35	0.25	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+78

Nomenclature: See separate page